Lecture 8

February 3， 2026

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Portfolio Construction

- Think of Portfolio construction taking three steps:
  1. Capital Allocation – weighting between risk-free assets and risky portfolio
  2. Asset Allocation – within risky portfolio, weighting of asset classes
  3. Security Selection – individual assets
- Asset Allocation and Security Selection technically the same
  - However, usually separated into two processes for practical reasons
  - Broad outlines of portfolio then make specific securities selected to fill outlines
  - Allow specialization within institution

Diversification

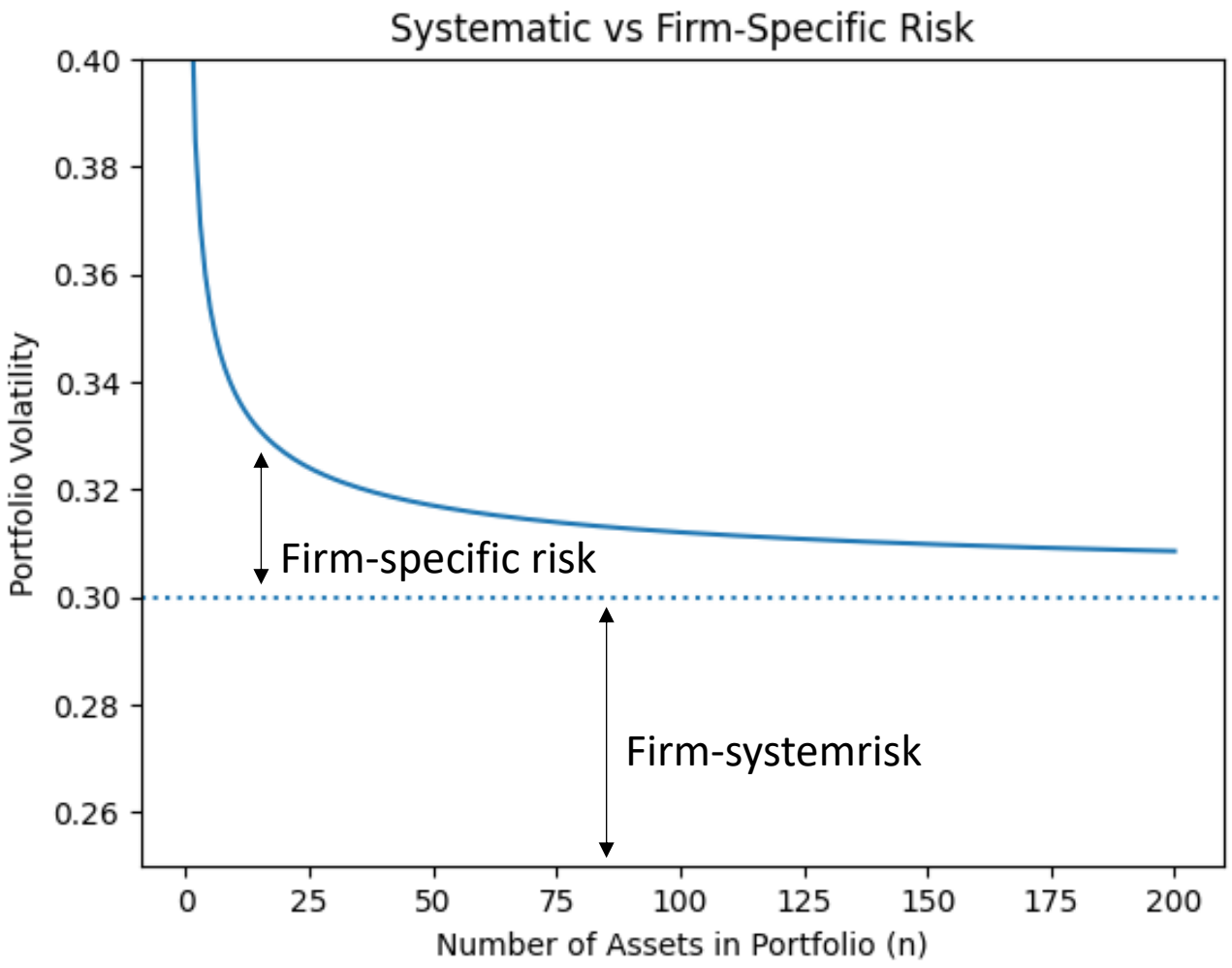
- Investors typically hold portfolios, not individual securities
- Risk of a portfolio depends not only on individual asset risk, but on how assets move together
- Diversification exploits imperfect correlation between asset returns
- By combining assets, investors can reduce total portfolio risk without proportionally reducing expected return
- Diversification is not about eliminating risk, but about eliminating uncompensated risk
- Sets the stage for distinguishing diversifiable (idiosyncratic) risk from non-diversifiable (systematic) risk

Types of Risk: Systematic vs Diversifiable

- Total risk of an asset can be decomposed into two components
  - Systematic (Market) Risk:
    - Risk related to broad economic and market-wide factors
    - Affects many or all assets simultaneously
    - Examples: recessions, interest rate changes, inflation shocks
    - Cannot be eliminated through diversification
    - Commands a risk premium in equilibrium
  - Idiosyncratic (Firm-Specific) Risk:
    - Risk unique to a particular firm or industry
    - Examples: management changes, product failures, lawsuits
    - Can be reduced or eliminated through diversification
    - Does NOT earn a risk premium in well-diversified portfolios
- Diversification works by eliminating idiosyncratic risk, not systematic risk

Types of Risk: Systematic vs Diversifiable (con.)

- As we add assets, portfolio volatility declines — but at a decreasing rate.
- However, volatility never goes to zero — it converges to a lower bound.
- Diversification reduces firm-specific risk, but market risk remains even in very large portfolios.



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Portfolio Return: Two Risky Assets

- Let:
  - w_1 = weight on asset 1
  - $w_2 = 1 - w_1$ = weight on asset 2
  - $w_1 + w_2 = 1$
  - X,Y = random returns
- Portfolio return (definition): $R_p = w_1 X + w_2 Y$
- So expected return:
  - $E(R_p) = w_1 E(X) + w_2 E(Y)$
- Expected return depends only on weights and expected returns — not on correlation.
  - This is why diversification does not reduce expected return.

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From Variance to Covariance (Joint Risk)

- Variance (single variable risk)
  - For a random variable X with mean μ_X:

$$\mathrm{Var}(X)=E[(X-\mu_X)^2]$$

- The Covariance between two data series (of random variables) is the extent to which they move together.
- Covariance (joint risk / co-movement):
  - For two random variables X,Y with means μ_X,μ_Y:

$$\mathrm{Cov}(X,Y)=E[(X-\mu_X)(Y-\mu_Y)]$$

  - Interpretation:
    - >0: returns tend to move together
    - <0: returns tend to move in opposite directions
    - =0: no linear co-movement
- Covariance is the object that makes diversification possible.

Covariance — Scenario / Probability Form

- If outcomes i=1,…,n occur with probabilities p_i:

$$\mathrm{Cov}(X,Y) = \sum_{i=1}^{n} (x_i - \mu_X)(y_i - \mu_Y)\, p_i$$

- Special case (equal probabilities $p_i = 1/n$):

$$\mathrm{Cov}(X,Y) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu_X)(y_i - \mu_Y)$$

- This shows covariance is the average product of joint deviations.

Correlation as a Scaled Covariance

- Definition of Correlation:

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

  - Correlation rescales covariance
  - Bounded between −1 and +1
  - Useful for intuition, not for optimization
- Rearranging:

$$\mathrm{Cov}(X, Y) = \rho_{XY}\, \sigma_X\, \sigma_Y$$

- This is the form used in portfolio variance calculations.

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Two-Asset Portfolio Variance (Using Covariance)

- Let portfolio return be:
  - $Rp=w1X+w2Y, \quad w1+w2=1$
- Start from the variance of a linear combination:
  - $Var(R_p)=Var(w_1X+w_2Y)$
- Apply variance rules:
  - $=w_1^2\,Var(X)+w_2^2\,Var(Y)+2w_1w_2Cov(X,Y)$
- Substitute covariance using correlation:
  - $\sigma_p^2=w_1^2\sigma_X^2+w_2^2\sigma_Y^2+2w_1w_2\rho_{XY}\sigma_X\sigma_Y$

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Portfolio Variance Example

- 3 states of the world:

State (i)	(p_i)	(x_i) (return on (X))	(y_i) (return on (Y))
1	0.25	−0.10	0.00
2	0.50	0.10	0.08
3	0.25	0.30	0.16

Compute means:

$$\mu_X = \sum p_i x_i = 0.10, \qquad \mu_Y = \sum p_i y_i = 0.08$$

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Portfolio Variance Example (con.)

- Compute variances and standard deviations:

$$\mathrm{Var}(X) = \sum p_i (x_i - \mu_X)^2 = 0.02, \quad \sigma_X = \sqrt{0.02} = 0.141421$$

$$\mathrm{Var}(Y) = \sum p_i (y_i - \mu_Y)^2 = 0.0032, \quad \sigma_Y = \sqrt{0.0032} = 0.056569$$

- Compute covariance:

$$\mathrm{Cov}(X, Y) = \sum p_i (x_i - \mu_X)(y_i - \mu_Y)$$

  - deviations for X: [−0.20,0,+0.20]
  - deviations for Y: [−0.08,0,+0.08]
  - products: [0.016,0,0.016]
  - probability-weighted products: [0.004,0,0.004]
  - So: Cov(X,Y)=0.004+0+0.004=0.008

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Portfolio Variance Example (con.)

- Correlation via scaling formula:

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{0.008}{(0.141421)(0.056569)} = 1.0$$

- Choose weights and compute portfolio risk
  - Let w_1=0.6, w_2=0.4, and $R_p=w_1X+w_2Y$
  - Expected return: $E(R_p)=w_1\mu_X+w_2\mu_Y$
    =0.6(0.10)+0.4(0.08)=0.092
  - Variance: $\sigma_p^2=w_1^2\sigma_X^2+w_2^2\sigma_Y^2+2w_1w_2\mathrm{Cov}(X,Y)$
    =0.36(0.02)+0.16(0.0032)+2(0.6)(0.4)(0.008)=0.011552
  - Volatility:

$$\sigma_p = \sqrt{0.011552} = 0.10748$$

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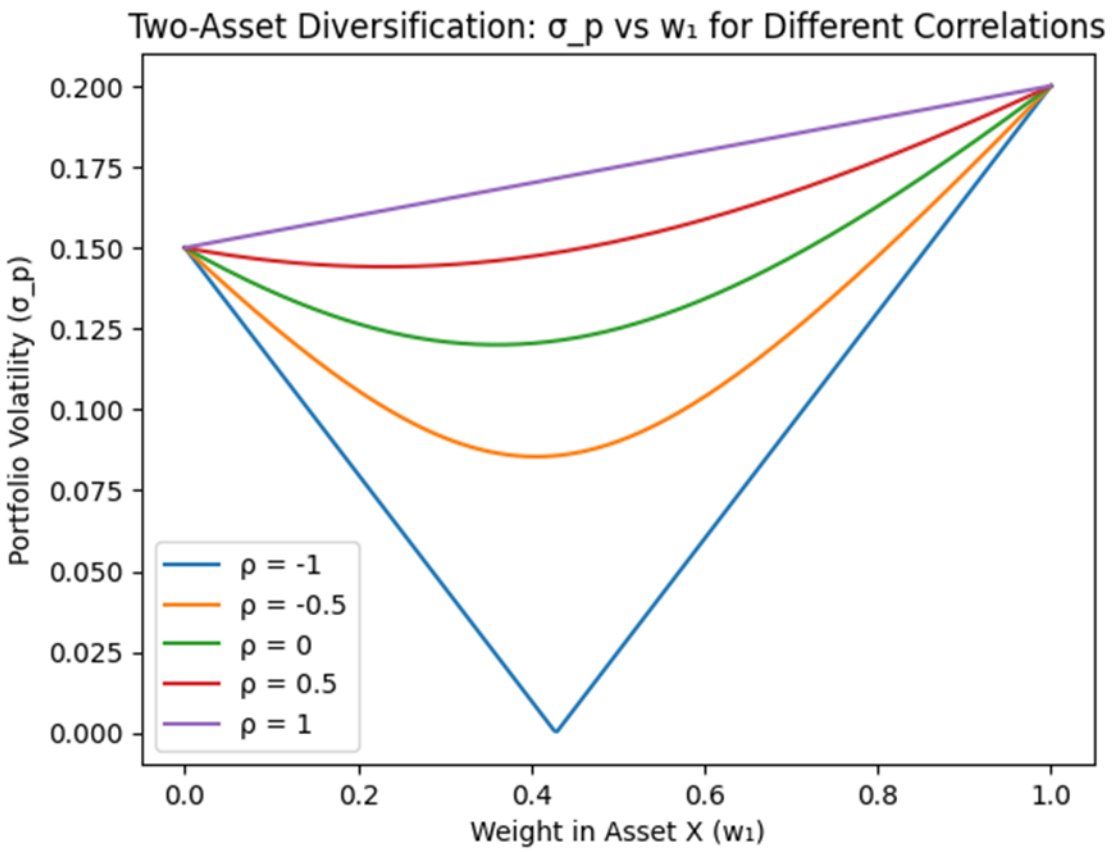
Diversification and the Role of Correlation

- Portfolio risk depends critically on the correlation between asset returns
- When returns move together, diversification provides little or no risk reduction
- As correlation falls, combining assets reduces portfolio volatility
- Negative correlation allows risk to be offset across assets



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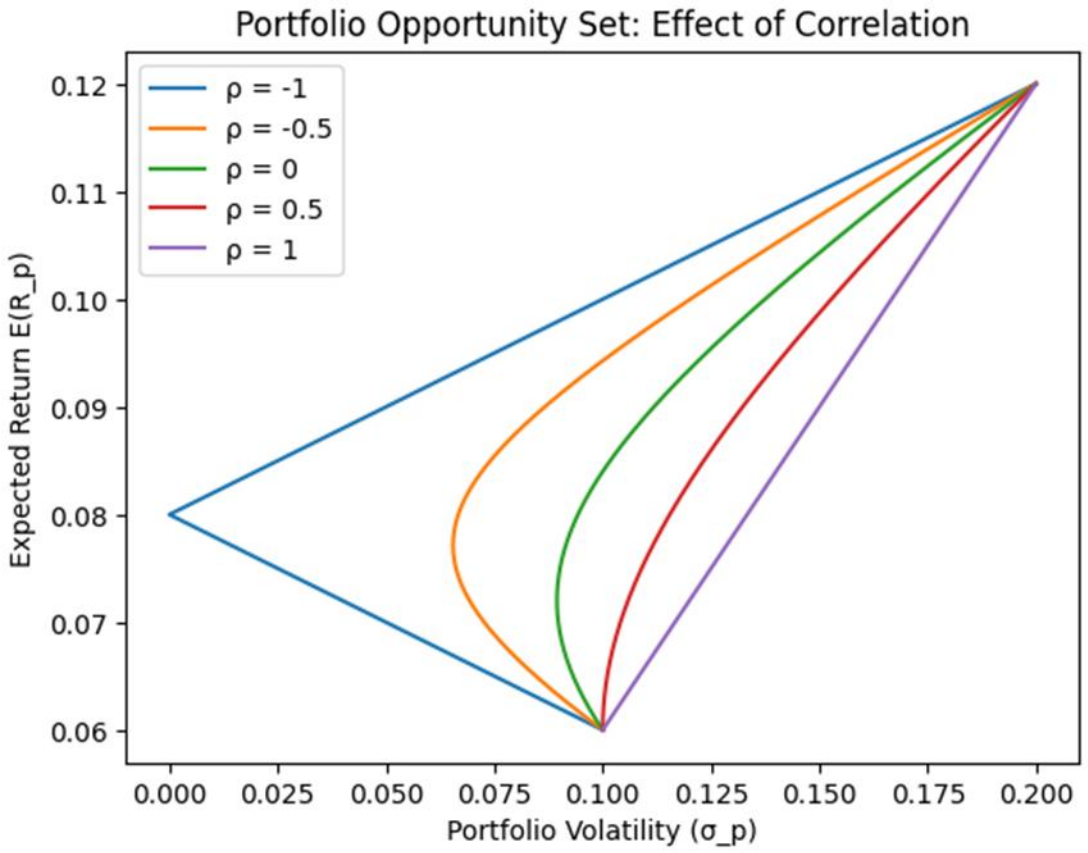
Portfolio Opportunity Set: The Role of Correlation

- Each curve represents the set of all feasible portfolios formed from two risky assets
- Curves differ only in the correlation ρ between asset returns
- Lower correlation shifts the opportunity set leftward, reducing risk for a given expected return
- When ρ=+1, diversification provides no risk reduction
- As ρ falls, diversification benefits increase and the opportunity set expands



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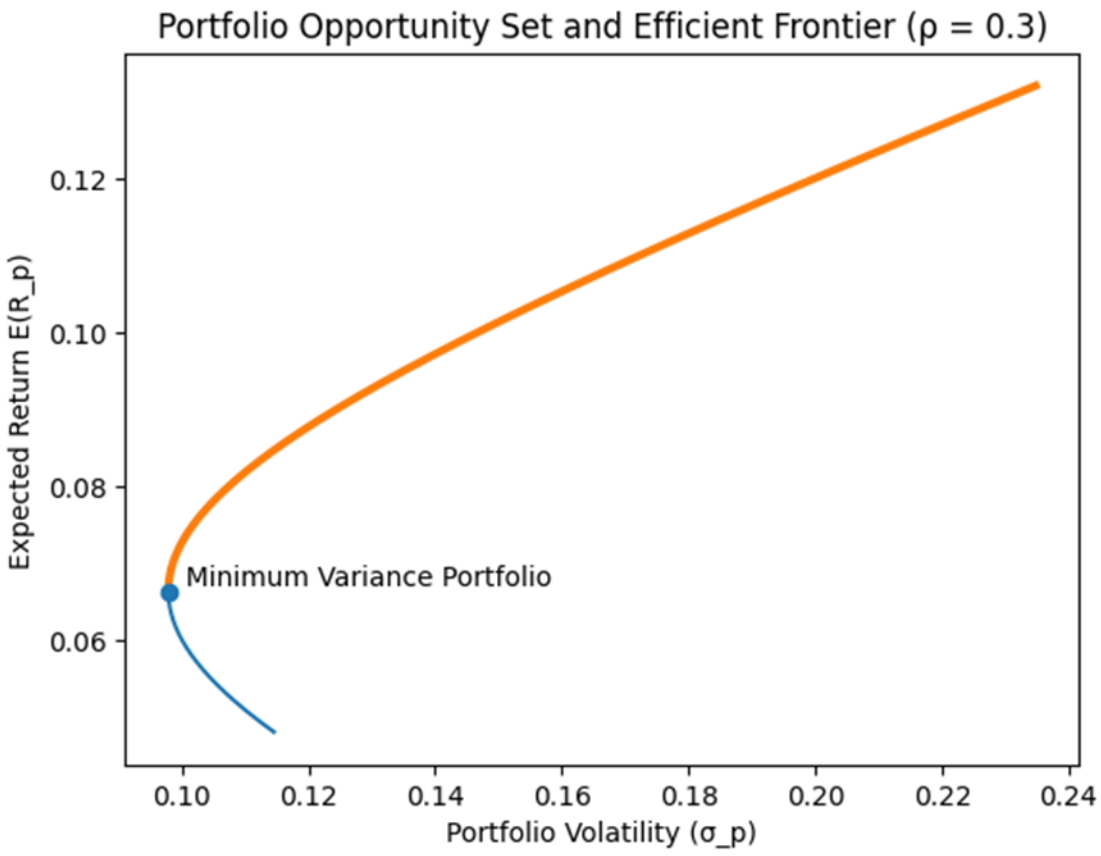
From Opportunity Sets to the Efficient Frontier

- The portfolio opportunity set contains all feasible portfolios
- Not all feasible portfolios are economically relevant
- Some portfolios are dominated by others:
  - same risk with lower expected return, or
  - same expected return with higher risk
- Investors therefore restrict attention to mean–variance efficient portfolios
- The efficient frontier is the set of portfolios offering the highest expected return for each level of risk

Portfolio Opportunity Set and Efficient Frontier

- The curve shows all feasible portfolios formed from two risky assets
- Each point represents a different allocation across the two assets
- The left-most point is the Minimum Variance Portfolio (MVP)
- Portfolios below the MVP are inefficient:
  - for the same risk, higher expected return is attainable
- Portfolios above the MVP form the efficient frontier
- Rational investors restrict attention to the efficient frontier



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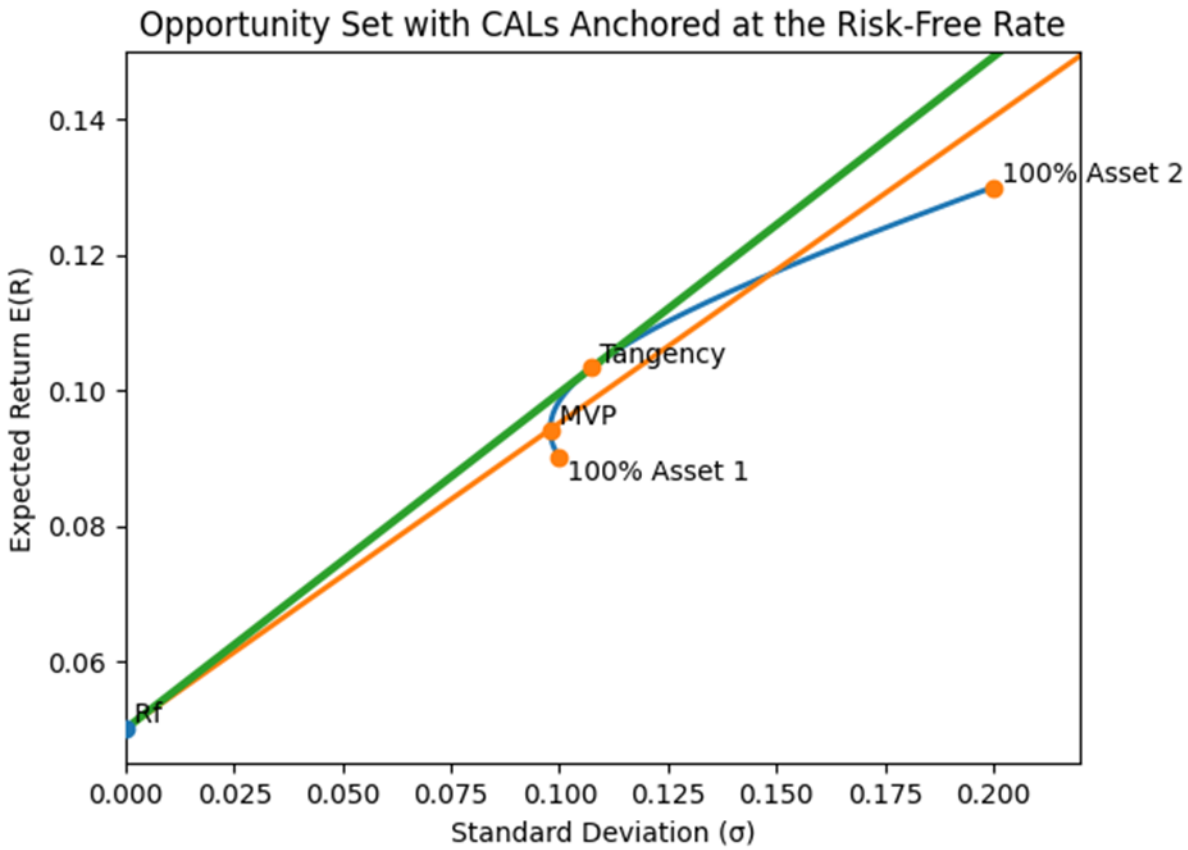
Opportunity Set with Alternative Capital Allocation Lines

- The curve shows the opportunity set of two risky asset classes
- Introducing a risk-free asset allows investors to form Capital Allocation Lines (CALs)
- Each CAL represents combinations of the risk-free asset and a risky portfolio
- All CALs must originate at the risk-free rate
- Different risky portfolios generate CALs with different slopes



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Sharpe Ratio Maximization

- The slope of a Capital Allocation Line measures its attractiveness
- Steeper CALs dominate flatter CALs for all investors
- The optimal risky portfolio is the one that generates the steepest CAL
- This portfolio maximizes expected excess return per unit of risk
- The optimal risky portfolio is the portfolio with the maximum Sharpe ratio

Tangency Condition for the Optimal Risky Portfolio

- Let excess returns be defined as:
  - $r_X \equiv R_X - R_f$, $r_Y \equiv R_Y - R_f$
- For the optimal risky (tangency) portfolio T, excess returns are proportional to covariance with the portfolio:

$$r_i \;\propto\; \mathrm{Cov}(R_i, R_T)$$

- For two risky assets X and Y, this implies the ratio condition:

$$\frac{E(r_X)}{E(r_Y)} = \frac{\mathrm{Cov}(R_X, R_T)}{\mathrm{Cov}(R_Y, R_T)}$$

- Define the risky portfolio

$$R_T = w_X R_X + w_Y R_Y, \qquad w_Y = 1 - w_X$$

  - This condition characterizes the tangency portfolio that maximizes the Sharpe ratio.

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Tangency Condition to the Optimal Risky Weight Formula

- Expand portfolio covariances
  - Using linearity of covariance:

$$\mathrm{Cov}(R_X, R_T) = w_X \sigma_X^2 + w_Y \,\mathrm{Cov}(R_X, R_Y)$$

$$\mathrm{Cov}(R_Y, R_T) = w_X \,\mathrm{Cov}(R_X, R_Y) + w_Y \sigma_Y^2$$

  - Let: $\sigma_{XY} \equiv \mathrm{Cov}(R_X, R_Y)$
- Substitute into the tangency condition:

$$\frac{E(r_X)}{E(r_Y)} = \frac{w_X \sigma_X^2 + (1 - w_X)\sigma_{XY}}{w_X \sigma_{XY} + (1 - w_X)\sigma_Y^2}$$

Tangency Condition to the Optimal Risky Weight Formula (con.)

- Cross-multiply:

$$E(r_X)\left[w_X\sigma_{XY} + (1 - w_X)\sigma_Y^2\right] = E(r_Y)\left[w_X\sigma_X^2 + (1 - w_X)\sigma_{XY}\right]$$

- Expand and collect w_X terms:

$$E(r_X)\sigma_Y^2 + w_X E(r_X)(\sigma_{XY} - \sigma_Y^2) = E(r_Y)\sigma_{XY} + w_X E(r_Y)(\sigma_X^2 - \sigma_{XY})$$

- Move constants to one side and w_X terms to the other:

$$w_X\Big(E(r_X)(\sigma_{XY} - \sigma_Y^2) - E(r_Y)(\sigma_X^2 - \sigma_{XY})\Big) = E(r_Y)\sigma_{XY} - E(r_X)\sigma_Y^2$$

Tangency Condition to the Optimal Risky Weight Formula (con.)

- Solve for w_X and simplify:

$$w_X = \frac{E(r_X)\sigma_Y^2 - E(r_Y)\sigma_{XY}}{E(r_X)\sigma_Y^2 + E(r_Y)\sigma_X^2 - \big(E(r_X) + E(r_Y)\big)\sigma_{XY}}$$

- and:

$$w_Y = 1 - w_X$$

- Where:

$$E(r_X) = E(R_X) - R_f,\ E(r_Y) = E(R_Y) - R_f$$
$$\sigma_{XY} = \mathrm{Cov}(R_X, R_Y)$$

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Tangency (Optimal Risky) Portfolio Example

- Given:
  - Risk-free rate: R_f = 5%
  - Risky Assets:
    - $E(R_X)$ = 9%, σ_X= 10%
    - $E(R_Y)$ = 13%, σ_Y= 20%
  - Correlation: ρ_{XY}=0.30
- Compute excess returns and covariance:

$$E(r_X) = 0.09 - 0.05 = 0.04, \qquad E(r_Y) = 0.13 - 0.05 = 0.08$$

$$\sigma_{XY} = \rho_{XY}\sigma_X\sigma_Y = 0.30(0.10)(0.20) = 0.006$$

Tangency (Optimal Risky) Portfolio Example (con.)

- Apply Optimal Risky Weight Formula:

$$w_X = \frac{E(r_X)\sigma_Y^2 - E(r_Y)\sigma_{XY}}{E(r_X)\sigma_Y^2 + E(r_Y)\sigma_X^2 - \left[E(r_X) + E(r_Y)\right]\sigma_{XY}}$$

$$w_X = \frac{0.04(0.04) - 0.08(0.006)}{0.04(0.04) + 0.08(0.01) - 0.12(0.006)} = \frac{0.00112}{0.00168} = \boxed{0.667}$$

$$\boxed{w_Y = 1 - w_X = 0.333}$$

- Optimal risky portfolio:
  - 66.7% Asset X
  - 33.3% Asset Y

From the Tangency Portfolio to Investor Choice

- The tangency portfolio maximizes the Sharpe ratio
- All investors hold the same optimal risky portfolio
- Investors differ only in how much risk they choose to take
- Remember Mean–variance utility:

$$U = E(R_C) - \frac{A}{2}\sigma_C^2$$

- where:
  - A= coefficient of risk aversion
  - C = complete portfolio (risky portfolio + risk-free asset)
- Optimal allocation to the risky portfolio:

$$y^* = \frac{E(r_T)}{A\sigma_T^2}$$

- where:
  - $E(r_T)$ =expected excess return on the tangency portfolio
  - σ_T^2 =variance of the tangency portfolio

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Optimal Complete Portfolio Example

- Tangency portfolio characteristics:
  - From the previous slide:

$$E(r_T) = w_X E(r_X) + w_Y E(r_Y)$$

$$E(r_T) = 0.667(0.04) + 0.333(0.08) = 0.0533$$

$$\sigma_T^2 = (0.667)^2(0.10)^2 + (0.333)^2(0.20)^2 + 2(0.667)(0.333)(0.006)$$

$$\sigma_T^2 \approx 0.0116$$

  - Investor A: Moderate risk aversion (A=4): $y^* = \frac{0.0533}{4(0.0116)} \approx 1.15$
    - 115% in risky portfolio vs −15% in risk-free asset (leveraged position)
  - Investor B: High risk aversion (A=8): $y^* = \frac{0.0533}{8(0.0116)} \approx 0.57$
    - 57% in risky portfolio vs 43% in risk-free asset

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Final Allocation Across All Assets

- Optimal allocation to the risky portfolio
  - From the previous slide: $y^* = \frac{E(r_T)}{A\sigma_T^2}$
  - y^* = fraction invested in the optimal risky portfolio T
  - $1-y*$ = fraction invested in the risk-free asset
- Decompose the risky portfolio
  - Recall the tangency (optimal risky) portfolio weights: w_X=0.667, w_Y=0.333
  - Thus, the complete portfolio allocates:

$$\begin{aligned} \text{Asset } X: &\quad y^* w_X \\ \text{Asset } Y: &\quad y^* w_Y \\ \text{Risk-free asset}: &\quad 1 - y^* \end{aligned}$$

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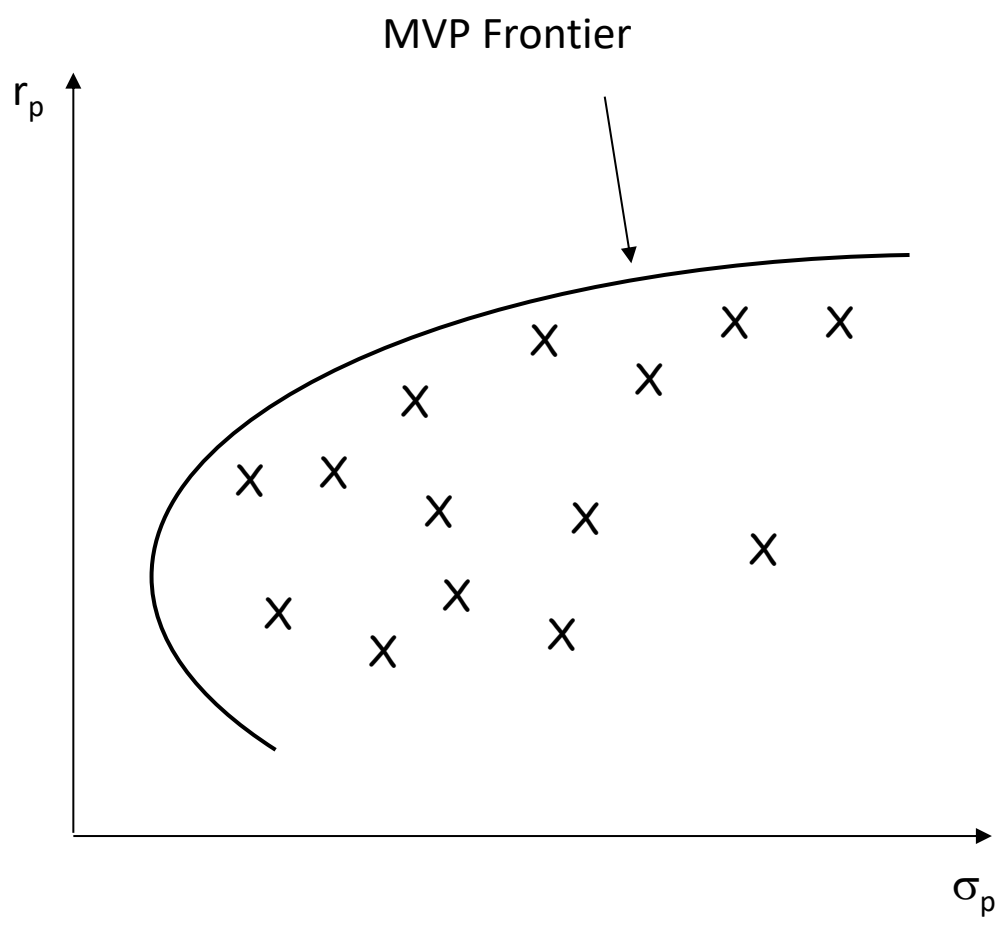
Final Allocation Across All Assets

- Investor A: Moderate risk aversion (A=4)
  - From earlier $y^* \approx 1.15$
    - Asset X: 1.15×0.667≈0.77(77%)
    - Asset Y: 1.15×0.333≈0.38(38%)
    - Risk-free asset:1−1.15=−0.15(−15%)
- Investor B: High risk aversion (A=8)
  - From earlier: $y^* \approx 0.57$
    - Asset X:0.57×0.667≈0.38(38%)
    - Asset Y:0.57×0.333≈0.19(19%)
    - Risk-free asset:1−0.57=0.43(43%)

Many Assets Represent the Minimum-Variance Frontier



- It can be shown (mathematically) that if we have hundreds or thousands of different assets, they will lie inside a hyperbola shaped like this.
- Notice that, assuming we allow short sales, none lie on the line.
- Shows us single asset portfolios are inefficient.
- Diversification allows us to build portfolios with higher expected return and lower σ

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The Optimal Asset Allocation Decision

- Step 1: Identify the universe of possible assets (including the riskless asset), and estimate returns, standard deviations, and all cross-correlations.
- Step 2: Identify the Efficient Frontier, i.e., the set of "efficient" portfolios that maximize return for each possible level of volatility (or equivalently, the portfolios that minimize volatility for each given level of returns).
- Step 3: Draw the steepest possible line (i.e., the tangent line) from r_f on the y-axis to the Efficient Frontier. This is the CML because it has the maximum possible return, at each risk level, that can be achieved using the universe of risky assets as well as the riskless asset.
- Step 4: Identify P, the optimal risky portfolio. P represents the optimal risky portfolio for all investors, regardless of their risk preferences
- Step 5: Assess the optimal asset allocation between the risky portfolio and the riskless asset for any given investor, based on their risk preferences (i.e., their minimum desired rate of return, or their maximum tolerated volatility). All investor portfolios will lie on the CML.

Practical Limitations of the Markowitz Mean–Variance Model

- Steps 1 & 2 are the Markowitz Model (Published in 1952, Nobel Prize in 1990)
- Quite the elegant solution; however, limitations:
  - Input Requirements - to implement Markowitz portfolio optimization, the investor must estimate:
    - Expected returns for all N assets: $E(R_1), E(R_2), \ldots, E(R_N)$
    - Variances and covariances among all assets: σ_{ij}=Cov(R_i,R_j), i,j=1,…,N
    - This requires estimating total parameter amounting to:

$$\underbrace{N}_{\text{means}} + \underbrace{\frac{N(N+1)}{2}}_{\text{variances \& covariances}}$$

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Practical Limitations of the Markowitz Mean–Variance Model (con.)

- Estimation Error:
  - Expected returns are not directly observable and must be estimated from historical data
  - Small errors in expected returns can lead to large changes in optimal portfolio weights
  - Mean–variance optimization is therefore highly sensitive to input assumptions
- This means optimal portfolios may be unstable and economically implausible.

Practical Limitations of the Markowitz Mean–Variance Model (con.)

- Computational Burden
  - As N grows, the variance–covariance matrix becomes large
  - Solving the optimization problem by hand is infeasible for realistic portfolios
- However:
  - The problem is quadratic and convex
  - It can be solved efficiently using:
    - spreadsheet solvers (e.g., Excel Solver)
    - statistical and numerical software (R, Python, MATLAB)
    - dedicated portfolio optimization tools

Practical Responses to Markowitz Limitations

- Use simplifying structure (e.g., index models, factor models) to reduce dimensionality
- Impose constraints (no short sales, bounds on weights) to stabilize solutions
  - How do you think constraints will affect Sharpe Ratio?
- Emphasize diversification and passive portfolios over finely tuned optimization
- Markowitz framework is theoretically elegant, but its practical implementation is limited by estimation error and data requirements rather than computational power.

From Markowitz to Index Models

- As we noted, Markowitz has its limitations, which can cause:
  - substantial estimation error
    - large long positions in some assets
    - large short positions in others
  - unstable and extreme portfolio weights
    - highly leveraged positions
    - concentrated portfolios
    - economically implausible allocations
    - large turnover
    - unstable rebalancing
    - high transaction costs
  - weak out-of-sample performance
    - Portfolios optimized on historical data often:
      - perform poorly on new data
      - exhibit lower realized Sharpe ratios than predicted

From Markowitz to Index Models

- Asset returns tend to co-move because they are driven by common risk factors
- Much of the covariance structure can be explained by exposure to these shared factors
- Index / Factor Model Approach
  - Decompose asset returns into:
    - a systematic (factor-driven) component
    - an idiosyncratic (asset-specific) component
  - Covariances arise primarily from common factor exposures
- Dramatically reduces the number of parameters to estimate
- Produces more stable and interpretable portfolio weights
- Preserves the diversification logic of Markowitz

The Separation Property and Practical Portfolio Management

- With a risk-free asset available, all investors hold:
  - the same optimal risky portfolio
  - combined with the risk-free asset in different proportions
- Investor risk preferences affect how much risk to take, not which risky assets to hold
- Portfolio choice separates into:
  - Step 1: Identify the optimal risky portfolio
  - Step 2: Allocate between the risky portfolio and the risk-free asset
  - [Essentially Steps 4&5 from The Optimal Asset Allocation Decision Slide]

The Separation Property and Practical Portfolio Management (con.)

- Implications for Investment Management
  - The separation property implies a natural division of labor:
    - Asset allocation determines exposure across asset classes
    - Security selection determines holdings within each asset class
  - Although technically part of the same optimization problem, these tasks are separated in practice due to:
    - scale and complexity of large portfolios
    - specialization in asset-specific information
    - organizational efficiency
- Institutional Implementation
  - Large investment firms typically separate responsibilities:
    - Asset Allocation Teams / CIO Office: determine overall risk exposure and asset class weights
    - Portfolio Managers / Analysts: select securities within a given asset class
  - This structure is consistent with mean–variance theory:
    - centralized determination of the risky portfolio
    - decentralized implementation within asset classes

The Power of Diversification

- Equal-Weighted Portfolio of N Risky Assets
  - Assume:
    - Each asset has variance σ^2
    - Pairwise correlation between assets is ρ
    - Portfolio weights are equal: w_i=1/N
  - The variance of the equally weighted portfolio is:

$$\sigma_p^2 = \frac{1}{N}\sigma^2 + \frac{N-1}{N}\rho\sigma^2$$

  - This expression separates portfolio risk into two components.

The Power of Diversification (con.)

- As $N \rightarrow \infty$ $$\lim_{N \to \infty} \sigma_p^2 = \rho\sigma^2$$
- The first term: $\frac{1}{N}\sigma^2 \rightarrow 0$
- The second term: $\frac{N-1}{N}\rho\sigma^2 \rightarrow \rho\sigma^2$
- Idiosyncratic (firm-specific) risk:
  - declines with portfolio size
  - diversified away as N increases
- Systematic (market) risk:
  - driven by correlation ρ
  - cannot be diversified away

Risk Pooling: Statistical Aggregation of Risk

- Risk pooling refers to combining independent or weakly correlated risks.
- Pooling is a statistical operation, not an allocation rule
- Variance of the average outcome declines with the number of pooled risks, but the total (gain/loss) variance increases.

Aggregate payoff variance:

$$\mathrm{Var}\left(\sum_{i=1}^{N} R_i\right) = N\sigma^2$$

Average payoff variance:

$$\mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N} R_i\right) = \frac{\sigma^2}{N}$$

Risk Sharing: Allocation of Pooled Risk

- Risk sharing determines how pooled risk is distributed across agents
- Achieved through:
  - ownership structures
  - markets
  - insurance contracts
- Efficient risk sharing assigns more risk to less risk-averse agents
- Institutional Examples
  - Insurance companies:
    - pool many independent risks
    - price and allocate risk via premiums
  - Financial markets:
    - distribute risk through asset ownership
    - allow heterogeneous risk tolerance